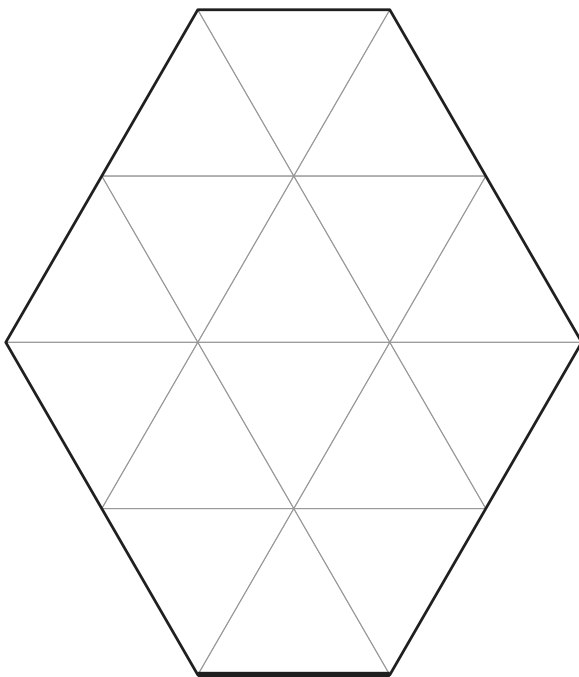
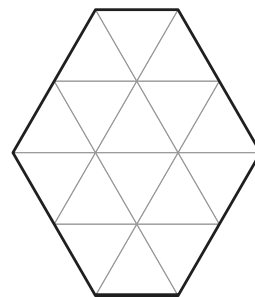
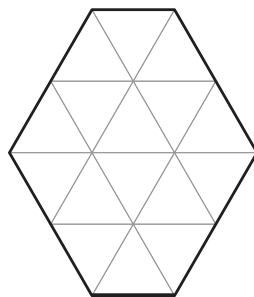
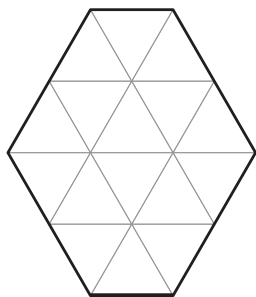
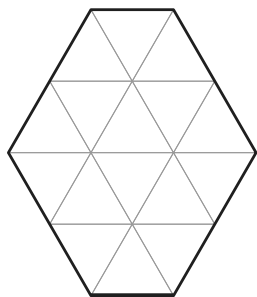
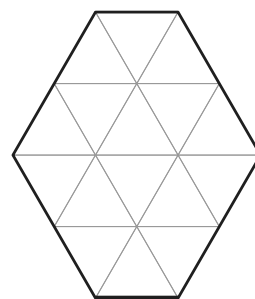
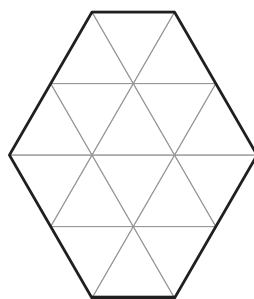
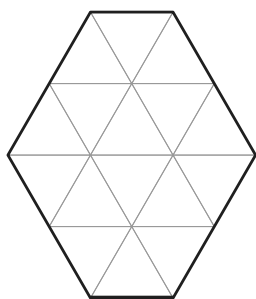
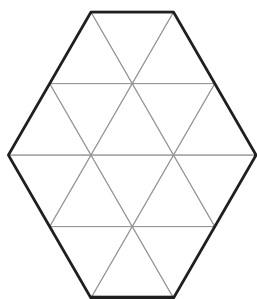


Problem 1: Fill the large board using only small blue rhombi. Find as many different tilings as you can. Keep the bold edge at the bottom. Record each tiling on a small board.

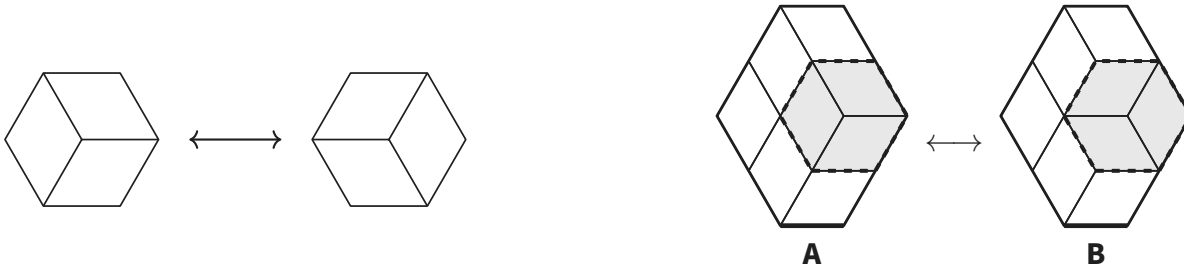


Record each blue rhombus with one line joining the centers of its two triangle spaces.

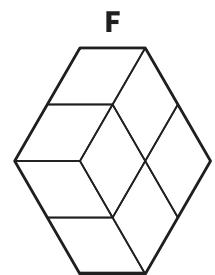
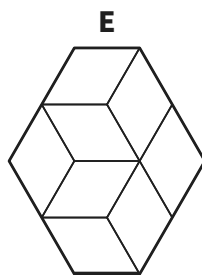
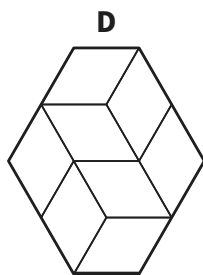
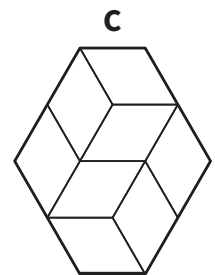
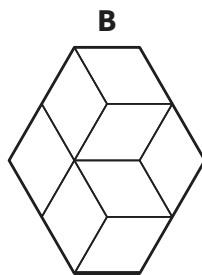
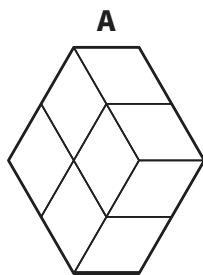


How many different tilings did you find? _____

Problem 2: A flip changes three blue rhombi inside a small hexagon to its other filling. The other rhombi stay in place. Try the flip shown from A to B, then flip back.



Make a map on these six cards. Draw a line between two cards whenever one flip changes one into the other. Check each line with your blocks.



Can you get from every card to every other card by a sequence of flips?

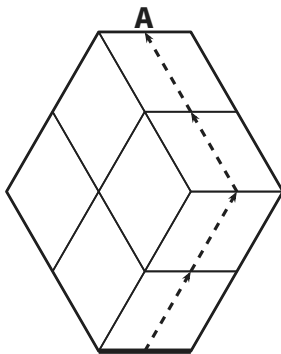
Problem 3: Find the shortest route from A to F. Write the cards you visit and count your flips.

Route: A _____

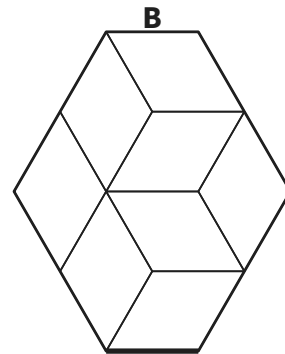
Number of flips: _____

Problem 4: Can you leave A and return to A in exactly three flips? Five flips? Any odd number of flips? Explain what you notice.

Problem 5: Follow a ribbon from the middle of the bold bottom edge. Cross its rhombus to the middle of its opposite horizontal edge. Keep crossing upward until you reach the top. Follow the example on A, then draw the ribbon on B. Write L for each step to the left and R for each step to the right, reading from bottom to top.



Code: _____



Code: _____

At any step, is there a choice of which rhombus to cross next? Could one tiling have two different ribbons starting at the bold edge?
